

Answer key

Version A

Marking guide and worked solutions · 40 points · **instructor copy: do not distribute before the exam**

How to mark. Sections 1 and 5 and the explanation parts of the others reward **sensible reasoning that is defended with a reason or a number**; there is no single right model. Apply **follow-through** from earlier answers. Accept rounding differences of ± 0.02 in s and t and ± 0.5 in forecasts. Students interpret coefficients but are **not yet** asked whether coefficients are significant. In Section 4 a t -test is **significant if $p < 0.05$** (two-tailed, T.TEST type 2). Objective column: syllabus learning outcome (LO) and your category, **Con** concept, **Cal** calculation, **Com** computation in Sheets.

1 · Quick answer sheet

Q	Objective	Expected answer	Pts	How the marks split
1a	LO1, LO5 · Con	Any three sensible inputs with a direction and a reason: temperature (up: hot days want cold treats), weekend (up: more leisure time), beach event (up: crowds nearby), planned price (down: dearer gelato sells less), sunshine (up, but overlaps with temperature).	3	1 per input: direction and a coherent reason. A “wrong” direction with a good reason scores; a direction with no reason does not.
1b	LO5 · Con	Item 6 , iced coffee sales counted at closing time: it is not known before opening (only after the day) and it is driven by the same weather and day type, so it cannot be used to forecast.	2	1 identifies item 6 + 1 the problem (not known in advance).
1c	LO3, LO5, LO6 · Con	Not both. With $r = 0.78$ they carry mostly the same information, so the model cannot tell their effects apart, and “holding sunshine constant while temperature changes” is unrealistic because they rise and fall together: the coefficients become unreliable (the model assumes inputs that are not duplicates of each other). Keep temperature ($r = 0.58$ with sales vs 0.49 ; a more direct measure of heat).	2	1 overlap and why it is a problem + 1 a choice with a reason.
1d	LO6 · Con	Hold back some days, fit the model on the rest, predict the held-back days, then calculate their residuals and MAE.	1	1: a hold-out check with residuals/MAE (names the idea of unseen days).
2a	LO5, LO6 · Com	Sales = $48.97 + 5.57 \times \text{temperature} + 34.16 \times \text{weekend} + 25.74 \times \text{event}$ ($\hat{y} = b + m_1x_1 + m_2x_2 + m_3x_3$; the table shows Col A = m_3 , Col B = m_2 , Col C = m_1 , Col D = b).	2	1 correct numbers matched to the right variables + 1 correct equation form.
2b	LO6, LO7 · Con	(i) 48.97 is the predicted sales when temperature = 0°C , it is a weekday and there is no event; it is not the sales on an average day (and 0°C is outside the data). (ii) On event days the model predicts about 26 more cups than on days without an event, holding temperature and weekend fixed; an association, not proof that events cause the sales. (iii) About 61% of the day-to-day variation in sales is explained by the model; a typical forecast misses by about 23 cups ($SE - Y$).	3	1 each: the corrected statement must keep units and “holding the others fixed”.
2c	LO3, LO6 · Cal + Con	(a) $5 \times 5.57 = 27.85$, about +28 cups . (b) The 5°C rise (+27.9) changes predicted sales slightly more than a beach event (+25.7). (c) The coefficients have different units (per $^\circ\text{C}$ and per event day): multiply each by the actual change in its variable before comparing.	3	1 + 1 + 1.
2d	LO5, LO6 · Cal	(a) $\hat{y} = 48.97 + 5.57(26) + 34.16(1) + 25.74(1) = \mathbf{253.7}$ cups. (b) Residual = $215 - 253.7 = \mathbf{-38.7}$. (c) $ \text{residual} = 38.7$ is larger than $SE - Y = 22.52$: a bigger miss than typical (about $1.7 \times SE - Y$): the café sold less than the model predicted; look for a reason (weather on the day, a closure, a stock-out).	4	2 prediction (1 substitution, 1 value) + 1 residual (actual – predicted) + 1 comparison with $SE - Y$.
3a-i	LO3, LO6 · Cal	Filled table: Mon actual 180, predicted 190, residual –10; Sat actual 245, predicted 230, residual +15; Wed actual 160, predicted 165, residual –5.	1	1 all blanks correct (follow-through on a single slip).
3a-ii	LO3, LO6 · Cal	$MAE = (10 + 15 + 5) \div 3 = \mathbf{10}$ cups.	1	1 correct (absolute values used).

3a-iii	LO6 · Con	Model Z has the lower MAE (6 vs 10). But every residual has the same sign: Z over-predicts every day by 6 (systematic bias). A simple correction (subtract 6) would remove it. Model X's errors are larger but not one-sided.	2	1 lower MAE with numbers + 1 identical signs = bias.
3b	LO3, LO4, LO6 · Con	(i) Day 8 : the hottest day, but sales far below the pattern. (ii) Without it the slope is 6.29 vs 2.49 with it, R^2 0.97 vs 0.17 (SE – Y 5.2 vs 28.8); one day cuts the slope by more than half and wrecks the fit. (iii) Find out why (closure, power cut, stock-out, recording error). Delete only with a documented reason; otherwise keep it and report results both ways.	4	1 day + 2 impact (1 slope, 1 R^2) + 1 sensible next step.
4a	LO3 · Cal	Means 200 and 186; $\Sigma(x - \bar{x})^2 = 320$ and 256; $s_1 = \sqrt{(320 \div 4)} = 8.94$, $s_2 = \sqrt{(256 \div 4)} = 8.00$ (deviations in the worked table below).	3	1 means + 1 sums of squares + 1 s values.
4b	LO3 · Cal	$s_p = \sqrt{[(4 \times 80 + 4 \times 64) \div 8]} = \sqrt{72} = 8.49$; $t = (200 - 186) \div [8.49 \times \sqrt{(1/5 + 1/5)}] = 14 \div 5.37 = \mathbf{2.61}$.	2	1 s_p + 1 t (follow-through on s; sign may be reported either way if direction is stated).
4c	LO1, LO3, LO6 · Con	(i) H_0 : the price rise made no difference to mean daily gelato sales (mean before = mean after). (ii) $p = 0.031 < 0.05$, so the difference is statistically significant : reject H_0 . (iii) If the price had no effect, a difference this large or larger would turn up in only about 3% of samples like these. It is not the probability that H_0 is true and it does not give the size of the effect.	3	1 null hypothesis + 1 decision using p + 1 correct plain-words meaning.
4d	LO6 · Con	Any sensible other change: temperature or season, a local event, tourist numbers, a menu or staff change; only 5 days per group; this shows association rather than proof of cause.	1	1 a relevant reason/idea.
5	LO7 · Con	(i) The inputs recommended (e.g. temperature, weekend and beach events all push sales up (not sunshine, which overlaps with temperature)) with the typical forecast error (about 23 cups). (ii) The price test: gelato sales were lower after the price rise (about 14 cups, $p = 0.031$). (iii) A caution (small samples, other changes, association not proof, forecasts can still miss by the typical error).	3	1 inputs + main LINEST finding in plain words + 1 price test result in plain words + 1 a genuine caution.

2 · Worked tables

Section 4 worksheet (Table 6).

Day	Before (x_1)	$x_1 - \bar{x}_1$	$(x_1 - \bar{x}_1)^2$	After (x_2)	$x_2 - \bar{x}_2$	$(x_2 - \bar{x}_2)^2$
1	196	-4	16	184	-2	4
2	204	+4	16	192	+6	36
3	188	-12	144	176	-10	100
4	212	+12	144	196	+10	100
5	200	0	0	182	-4	16
Sum	1000	0	320	930	0	256
Mean $\bar{x}$	200			186		
s	8.94			8.00		

Check against Sheets: =T.TEST gives $p = 0.0312$ ($df = 8$, $|t| = 2.609$, two-tailed).

3 · Where students go wrong

- **Reading the LINEST table backwards** (2a): Col A is the **last** X column and Col D the intercept. The layout table is printed on the paper, so give no credit for guessing the order; apply follow-through if the order is wrong but the method is right.
- **“Per customer” and “causes”** (2b): the coefficients describe the **daily total** and an **association**.
- **Reading R^2 as “percent correct”** (2b): R^2 is the share of variation explained.
- **Using absolute values** (3a-ii): MAE averages the sizes of the errors, so signs must not cancel.
- **Reading p as the probability that H_0 is true** (4c). Credit “if there were no real effect, a result this extreme would be rare”.